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MIL-C-82747(OS)
27 January 1988

MILITARY SPECIFICATION

COATING, POLYURETHANE CLEAR, SAFFLOWER OIL MODIFIED ALKYD

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a white air-drying, safflower oil, modified polyurethane alkyd resin coating (see 6.1).

1.2 Classification. The coating shall be of the following two types:

Type I	Coating packaged in open cans
Type II	Coating packaged in aerosol cans

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

TT-T-291	Thinner, Paint, Volatile Spirits (Petroleum-Spirits)
PPP-P-1892	Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing
FED-STD-313	Material Safety Data Sheets, Preparation and Submission of
FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC: N/A

FSC 8010

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Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

Department of Labor

Code of Federal Regulations, Title 29
Part 1910, Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

PUBLICATIONS

American Society for Testing and Materials (ASTM)

C 346	Standard Test Method for 45-Deg Specular Gloss of Ceramic Materials
D 56	Standard Test Method for Flash Point by Tag Closed Tester (DoD adopted)
D 185	Standard Test Method for Coarse Particles in Pigments, Pastes and Paints (DoD adopted)
D 563	Standard Test Method for Phthalic Anhydride Content of Alkyd Resins and Resin Solutions (DoD adopted)
D 609	Standard Test Methods for Preparation of Steel Panels for Testing Paint, Varnish, Lacquer, and Related Products (DoD adopted)
D 968	Standard Test Methods for Abrasion Resistance of Organic Coatings by the Falling Abrasive Tester (DoD adopted)
D 1210	Standard Test Methods for Fineness of Dispersion of Pigment Vehicle Systems (DoD adopted)
D 1308	Standard Test Methods for Effect of Household Chemicals on Clear and Pigmented Organic Finishes (DoD adopted)

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D 1475	Standard Test Method for Paint, Varnish, Lacquer and Related Products, Density of (DoD adopted)
D 1640	Standard Test Methods for Drying, Curing or Film Formation of Organic Coatings at Room Temperature (DoD adopted)
D 1737	Standard Test Methods for Elongation of Attached Organic Coatings with Cylindrical Mandrel Apparatus (DoD adopted)
D 2372	Standard Method of Separation of Vehicle From Solvent-Reducible Paints
D 2697	Standard Test Method for Volume Nonvolatile Matter in Clear or Pigmented Coatings (DoD adopted)
D 2794	Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
D 3335	Standard Test Method for Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy (DoD adopted)
D 3359	Standard Methods for Adhesion by Tape Test Measuring (DoD adopted)
D 3363	Standard Test Method for Film Hardness by Pencil Test

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa. 19103.)

Chemical Specialties Manufacturers Association (CSMA) Publication

CSMA Aerosol Guide	Method for Pressure Measurement in Metal Aerosol Containers

(Application for copies should be addressed to Chemical Specialties Manufacturers Association, Inc., 1001 Connecticut Avenue NW, Washington, DC 20036.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The coating material, for both type I and type II, shall consist of a safflower oil modified polyurethane alkyd resin, thinners conforming to TT-T-291, dryers and antiskinning agents as required to meet the requirements of this specification.

3.2 Qualitative requirements.

3.2.1 Condition in container. The coating material after premixing the pigment dispersion shall be a homogeneous pigmented liquid free from sedimentation, thickening and deterioration.

3.2.2 Appearance and color. The coating material shall be homogeneous and shall match white color chip, number 17875 conforming to FED-STD-595.

3.2.3 Chemical resistance. The dried film shall show no discoloration, change in gloss, blistering, softening, swelling, loss of adhesion or other visible defects.

3.2.4 Dilution stability. When thinned as specified, the coating shall remain stable and uniform, showing no precipitation, curdling or separation.

3.2.5 Odor. The odor of the coating material in the container or during application shall not be noxious.

3.2.6 Working properties. The coating material shall be a freely working product with acceptable leveling properties, that is, no seeds, runs, streaks or brush marks.

3.2.7 Type II propellant. The propellant shall be a hydrocarbon or a blend of hydrocarbons and halogenated hydrocarbons as required to assure that the coating conforms to the requirements as specified herein. When tested as specified in 4.5.1.2.1, neither vinyl chloride nor chlorofluorocarbons 11, 12, 114, or 115 shall be present.

3.2.8 Type II dispenser.

3.2.8.1 Container. The container shall be a commercial type metal pressure container, generally known as an aerosol container, of such construction as to assure acceptance of the finished package by common carriers operating in interstate commerce. The capacity of the container shall be sufficient to contain the specified net contents of the product with outage in accordance with generally recognized safe commercial practice. Each container shall be labeled as stated in 5.3.2.

3.2.8.2 Dispensing valve. The valve shall have a spray head which can be removed without releasing pressure from the aerosol. The valve shall have a metering orifice accessible to cleaning with a common pin after removal of the spray head. The removable spray head shall contain orifices of such dimensions as to produce spraying properties as required in 3.2.8.5. The lower end of the dip tube shall have a maximum clearance of 9.5 millimeters (3/8 inch) from the bottom of the rim of the can after storage for one week in the product (which allows for swelling of the tube).

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3.2.8.3 Agitation. Each dispenser of the pigmented material shall contain one or more agitators capable of dispersing the pigment by normal shaking. The agitators shall be made of materials which shall not react with or contaminate the coating or container.

3.2.8.4 Cover cap. The valve shall be protected from accidental functioning and damage by a rigid cover cap which shall be removable and replaceable. The color of the cover cap shall be an approximate match to the color of the coating in the container.

3.2.8.5 Spraying properties. When tested as specified in 4.5.2.2.4, the contents of the aerosol container shall be sprayable when stabilized at temperatures from 15° to 32° Celsius (°C) (60° to 90° Fahrenheit (°F)). A minimum of 98 percent by weight of contents shall be sprayable in a pattern such as to deposit a uniform coating on a smooth vertical surface. The film shall dry free of orange peel, sags, specks, or any other surface irregularities. Leftover coating of more than 2 percent shall constitute a defect.

3.3 Quantitative requirements. The quantitative requirements for both type I and type II shall be as specified in table I.

3.3.1 Type II quantitative requirements..

3.3.1.1 Net contents. Unless otherwise specified in the contract or purchase order (see 6.2 and 6.6), each can shall contain a minimum of 360 and a maximum of 454 grams (13-16 avoir. ounces) net weight of contents at the discretion of the contractor, and the unit shall be considered a nominal one-pint can. The contents shall meet the total solids specified in the appropriate column of table II which requires that each can, regardless of net contents, shall have the same minimum amount of solids per can.

3.3.1.2 Pressure. When tested as specified in 4.5.2.2.3, the aerosol containers shall not burst or show evidence of leakage, distortion, or any other defects at an equilibrium pressure which shall not exceed 140 pounds per square inch gauge (psig) at 55° ± 1°C (130° ± 2°F). Any failure of this test shall be cause for rejection of the batch.

3.4 Toxicity. The material shall have no adverse effect on the health of personnel when used for its intended purpose in accordance with the Material Safety Data Sheet (MSDS). Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service which will act as an advisor to the contracting activity.

3.5 Material Safety Data Sheet (MSDS). The contracting activity shall be provided an MSDS at the time of contract award. The MSDS shall be prepared in accordance with the requirements of 29 CFR 1910.1200(g), Hazard Communication, and included with each shipment. OSHA Form 174 MSDS or a more effective contractor developed MSDS format may be used.

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TABLE I. Quantitative requirements for type I and type II.

Characteristics	Minimum	Maximum	Test Paragraph
Abrasion resistance (liters/mil)	17	—	4.5.2.1.1
Adhesion, dry film	4	—	4.5.2.1.2
Density (pounds per gallon)	9.10	9.40	4.5.2.1.3
Drying time (air dry, 3-mil wet film, 50% relative humidity, 77°F)			4.5.2.1.4
dry-to-touch, minutes	—	30	
set-to-touch, hours	—	2	
hard-dry, hours	—	12	
Film hardness, pencil	2H	—	4.5.2.1.5
Fineness of grind	—	7	4.5.2.1.6
Flash point, °F	110	—	4.5.2.1.7
Flexibility [1/8-inch (3-mm) mandrel], cracks or chips	—	0	4.5.2.1.8
Forty-five degree specular gloss	87	—	4.5.2.1.9
Impact, in-lbs	80	—	4.5.2.1.10
Lead, percent	—	0.005	4.5.2.1.11
Leveling index	5	—	4.5.2.1.12
Non-volatile, percent by volume	43	43	4.5.2.1.13
Non-volatile, percent by weight	59	61	4.5.2.1.13
Phthalic anhydride, weight percent in alkyd vehicle solids	0.04	0.10	4.5.2.1.14
Pigment, volume percent in total non-volatile	16	18	4.5.2.1.15
Pigment, weight percent in total non-volatile	43	45	4.5.2.1.15
Sag resistance index	7.0	—	4.5.2.1.16
Skinning (48 hours in 3/4-inch container)	no skin	—	4.5.2.1.17
Vehicle, total volume percent in total non-volatile	69	71	4.5.2.1.18
Vehicle, total weight percent in total non-volatile	47	51	4.5.2.1.18
Viscosity, Brookfield	475	525	4.5.2.1.19

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Toxicological formulations. The contractor shall have the toxicological formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use (see 6.3 and 6.4).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), inspection shall be at a temperature of 25° ± 1°C (77° ± 2°F) and a relative humidity of 50 ± 5 percent.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified herein.

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4.4.1 Lot size. Lot size shall be as specified in the contract (see 6.2).

4.4.2 Sampling. For sampling purposes lot size for type I shall be expressed in gallons and lot size for type II shall be expressed in number of aerosol containers. Samples shall be taken in accordance with MIL-STD-105, using inspection level S-2 and an acceptable quality level (AQL) of 2.5.

4.5 Test methods.

4.5.1 Qualitative test methods.

4.5.1.1 Test methods for both types (I and II).

4.5.1.1.1 Condition in container. Condition in container shall be determined by visual examination with the use of a spatula or paddle, after transferring the material to a clear glass container.

4.5.1.1.2 Appearance and color. Appearance and color shall be determined by visual comparison of a dried film with a 3" × 5" white color chip conforming to number 17875 FED-STD-595.

4.5.1.1.3 Chemical resistance. The chemical resistance shall be determined in accordance with ASTM D 1308.

4.5.1.1.4 Diluent concentration. The diluent concentration shall be determined in accordance with FED-STD-141, Method 4203.

4.5.1.1.5 Odor. The odor shall be determined in the work space to be used for application of the coating.

4.5.1.1.6 Working properties. The working properties shall be determined by visual examination during the preparation of test panels for 4.5.1.1.2.

4.5.1.2 Test methods for type II only. For qualitative tests, the manufacturer may use the coating concentrate as it exists before packaging and pressurization, provided it is adjusted as specified in 4.5.2.2.1.

4.5.1.2.1 Analysis of propellant gas for chlorofluorocarbons and vinyl chloride (type II).

4.5.1.2.1.1 Special apparatus and materials. The chemical analysis described herein requires the following apparatus and materials:

a. The can-piercing, pressure-reassuring device recommended in the CSMA Aerosol Guide for determining the internal pressure of aerosol products. The device to be used is Model CBA-6 from Builders Sheet Metal Works, Inc. or equivalent.
b. A temperature-programmable gas chromatograph (Hewlett-Packard Model 5750 gas chromatograph, or equivalent) equipped with a 0.5-ml gas sampling valve and thermal conductivity detector.
c. Two stainless-steel gas chromatograph columns, 1.83 meters long by 3 mm outside diameter, packed with 60-200-mesh silica gel.
d. Standard propellant gases in lecture bottles (available from Matheson Gas Products, or other contractors), chlorofluorocarbon no. 11, 12, 114, 115, and vinyl chloride.
e. Lecture bottle control valve (Matheson Model 58, or equivalent).

4.5.1.2.1.2 Gas chromatographic analysis. Condition the chromatographic columns at 200 °C with helium as the carrier gas at a flow rate of 75-100 ml/minute for 1 hour. Remove the actuator and cap of an unused aerosol container. Puncture the can using the pressure measuring device and connect the device to the gas chromatograph. Triplicate chromatographic analyses shall be done on the aerosol propellant sample and on mixtures of the aerosol propellant with each of the five standard propellant gases. For mixing each standard propellant, with

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the aerosol propellant sample, use a "T" connection (Swagelock or equivalent). The "T" connection shall be removed entirely when the aerosol propellant sample alone is to be analyzed. Bubble each gas sample through the sampling valve at a moderate rate for 2 minutes before each run is made to purge the valve of any other gases. Analyze the samples using a temperature program from 100° to 200°C at 6°C/minute, with a 2 minute post injection hold at 100°C and a 5 minute upper limit hold at 200°C. There shall be no material in the aerosol propellant sample having the same retention time as any of the five standard propellant gases.

4.5.1.2.2 Dispenser. The dispenser and its components shall be inspected and evaluated for compliance with 3.2.8.

4.5.2 Quantitative test methods.

4.5.2.1 Test methods for both types (I and II).

4.5.2.1.1 Abrasion resistance. Abrasion resistance shall be measured in accordance with ASTM D 968.

4.5.2.1.2 Adhesion. Apply a film of typical application thickness to a cleaned metal panel and dry at room temperature for 24 hours. Test the adhesion, following the procedure of ASTM D 3359, by cross-hatching a suitable area of the film so that strips of the metal substrate appear.

4.5.2.1.3 Density. The density shall be determined in accordance with ASTM D 1475.

4.5.2.1.4 Drying time. Drying times (dry-to-touch, set-to-touch and hard-dry) shall be determined in accordance with ASTM D 1640.

4.5.2.1.5 Film hardness. Film hardness shall be determined in accordance with ASTM D 3363.

4.5.2.1.6 Fineness of grind. Fineness of grind shall be determined in accordance with ASTM D 1210.

4.5.2.1.7 Flash point. Flash point shall be determined in accordance with ASTM D 56.

4.5.2.1.8 Flexibility. Flexibility shall be determined by bending a cured film, formed in accordance with ASTM D 1737, around a 1/8-inch (3-mm) mandrel. Formation of cracks or chips shall cause rejection of the lot represented by the sample.

4.5.2.1.9 Forty-five degree specular gloss. Forty-five degree specular gloss shall be determined on the cured film in accordance with ASTM C 346.

4.5.2.1.10 Impact resistance. Impact resistance shall be measured in accordance with ASTM D 2794.

4.5.2.1.11 Lead content. Lead content shall be determined in accordance with ASTM D 3335.

4.5.2.1.12 Leveling properties. The panel shall be prepared by drawing down a film of the coating on clear polished plate glass with a New York Society for Paint Technology (NYPC) leveling blade. Allow the film to dry in a horizontal position for 24 hours and then determine the leveling index as follows:

Condition	Index
All pairs open, none flowed together	0
Pair A flowed to just contact; pairs B, C, D, E open	1
Pair A flowed together; pairs B, C, D, E open	2
Pair A flowed together; pair B flowed to contact; pairs C, D, E open	3
Pairs A and B flowed together; pairs C, D, E open	4
Pairs A and B flowed together; pair C flowed to contact; pairs D and E open	5

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Pairs A, B and C flowed together; pairs D and E open	6
Pairs A, B and C flowed together; pair D to just contact; pair E open	7
Pairs A, B, C and D flowed together; pair E open	8
Pairs A, B, C and D flowed together; pair E to just contact	9
Pairs A, B, C, D and E flowed together	10

4.5.2.1.13 Non-volatile contents by volume and by weight. The non-volatile contents percent by volume and percent by weight shall be determined in accordance with ASTM D 2697.

4.5.2.1.14 Phthalic anhydride content. The phthalic anhydride content shall be determined in accordance with ASTM D 563.

4.5.2.1.15 Pigment contents by volume and by weight. The pigment contents percent by volume and percent by weight shall be determined in accordance with ASTM D 2697.

4.5.2.1.16 Sag resistance. Mount a sealed leneta test chart on a suitable flat applicator. Set a leneta antisag meter at the top of the test chart with the open side of the blade facing the operator. Place a suitable quantity of coating directly in front of the blade and draw-down the coating. The completed draw-down shall then be placed in a vertical position with the stripes horizontal, the thinnest stripe on top. Allow to dry in this position and then determine the anti-sag index as follows: the lowest (i.e. heaviest) film thickness which does not touch the next lower stripe is the index stripe. Fractional values are obtained by adding to the index value a fractional value based on the degree to which the stripe below the index stripe has merged with the next stripe, as follows:

Degree of Merger	Add
Complete (intervening block is completely wet)	0.0
Not complete, but definitely more than half	0.2
Approximately half	0.4
Appreciable, but definitely less than half	0.6
Slight, just touching	0.8

4.5.2.1.17 Skinning. Lack of skinning shall be determined in accordance with ASTM D 185.

4.5.2.1.18 Vehicle contents. Vehicle content percent by volume and percent by weight shall be determined in accordance with ASTM D 2372.

4.5.2.1.19 Viscosity. The viscosity shall be determined with a Brookfield viscometer Model LVT using spindle #2 at 30 revolutions per minute and 25°C.

4.5.2.2 Test methods for type II only.

4.5.2.2.1 Test concentrate. With the container well agitated and in an upright position, pierce the top with a needle point, making a very fine hole which will allow the propellant to escape (conditioning at low temperature for 1 hour will facilitate this). After a few minutes, the pierced hole can be enlarged, and as soon as there is no further evidence of gas escaping, the top of the container can be cut open. Allow to stand until the contents are at room temperature, then transfer the contents quantitatively into a tared beaker using several washes of

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acetone. Evaporate on a water bath until no more propellant is liberated and until the contents are less than 40 percent of the original weight of the contents including propellant. Then readjust to exactly the following percent of original weight with n-butyl acetate:

Nominal can contents (avoirdupois ounces)	Percent
16	40.0
15	42.6
14	45.7
13	49.2

4.5.2.2.2 Total net contents. The total net contents shall be determined by calculating the difference in weight between the filled container and the completely empty one (see 4.5.2.2.1 for procedure for emptying). Wash out interior of can with acetone and dry. (Note that the weight of the agitators must be included in the tare.) Using the completely emptied container as a tare, additional checks can be made on filled containers. At least two containers shall be checked from each lot sampled since variation in weight of empty cans can be found. A check on full and empty cans shall be made on the same cans and conformance with 3.3.1.1 determined.

4.5.2.2.3 Pressure. Containers filled for shipment shall be submerged in a water bath maintained at 55° ± 1°C (130° ± 2°F). The pressure shall be measured after 15 minutes of immersion in accordance with the method described in the CSMA Aerosol Guide, "Methods for Pressure Measurement in Metal Aerosol Containers", method 2. Evaluate for conformance with the requirements specified in 3.3.1.2. Inspection level S-3 shall be used for the pressure test. Failure of any one sample to meet this test shall be cause for rejection of the lot.

4.5.2.2.4 Spraying properties. Spray a 10- by 30-centimeter (cm) (4- by 12-inch) steel panel prepared in accordance with ASTM method 609, Type I, method 5B. Set the panel at an angle of approximately 15 degrees from the vertical. Hold the dispenser in an upright parallel position to the panel at a distance of 25 to 30 cm (10 to 12 inches) and with the valve fully open, move the dispenser horizontally from one side of the panel to the other using a stroking motion. Spray in overlapping horizontal strokes from the top to the bottom of the panel. Apply two thin coats (each coat approximately 0.012 mm (0.005 inch) dried film thickness) one after the other. The second thin coat can be applied by spraying vertically in overlapping strokes from side to side. After application of the second thin coat, place the panel in a vertical position to dry for 5 minutes, and examine the film for smoothness of flowout, freedom from specks, bubbles, craters or any apparent abnormalities. Continue spraying the contents to completion on a flat surface and note the spraying properties at intervals. Calculate the leftover coating as follows:

$$\text{Percent of leftover coating} = \frac{C}{A - B} \times 100$$

where

A = weight in grams of full can
B = weight in grams of empty can
C = weight in grams of leftover coating

4.5.3 Lot acceptance or rejection. The lot represented by the sample shall be accepted when the tests in section 4 show that the sample meets all the requirements of section 3. Otherwise, the lot represented by the sample shall be rejected.

4.5.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defectives shall be scored in accordance with table II. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operation. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery.

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The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 percent defective.

4.5.4.1 Examination for defects. Examination for defects shall be as outlined in table II.

Table II. Classification of preparation for delivery defects.

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.
Contents (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

5. PACKAGING

5.1 Type I packaging, packing and marking. The coating shall be packaged, packed and marked in accordance with PPP-P-1892. The coating shall be furnished in 1-gallon or 20-liter containers.

5.2 Type II packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Twelve dispensers of coating of the same description, packaged as specified in 5.2, shall be packed in fiberboard boxes made from weather-resistant fiberboard with a bursting test strength of not less than 275 pounds per square inch. Fiberboard separators the full height of the box shall be required. The box flaps shall be secured with water-resistant adhesive applied to not less than 75 percent of the surface area of the contact between the flaps, or with 3-inch-wide waterproof tape applied to the full length of the seams and extending over the ends not less than 3 inches. Alternatively, wirebound, cleated plywood, or nailed wood boxes shall be acceptable shipping containers when lined with a waterproof barrier material. The barrier material shall be sealed at the edges with waterproof tape or adhesive.

5.2.2 Commercial. The coating, packaged as specified in 5.2, shall be packed in fiberboard boxes to ensure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under the National Motor Freight Classification or Uniform Freight Classification Rules.

5.3 Marking.

5.3.1 Normal marking. In addition to the marking required by contract or purchase order (see 6.2), containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.2 Special marking (type II). Legible commercial marking shall be acceptable. Marking shall be lithographed on the unit container or printed on a label securely affixed to the container and shall include the following:

a. Color number of FED-STD-595
b. Net weight in ounces
c. Name of material
d. Directions for including surface preparation
e. A formula in general terms
f. "For best results use at temperatures between 16° and 32° C (60° and 90° F)"
g. Date of manufacture
h. "Do not puncture or burn"
i. "Keep from fire or flame"

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j. "Use with adequate ventilation"
k. "Avoid repeated contact with skin"
l. "Store below 49° C (120° F)"
m. "Keep from children"
n. "Shake for 60 seconds before use"

6. NOTES

6.1. Intended use. The coating described herein is intended for use on the Standard Missile.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Type and size of container
c. Place of delivery
d. Whether net contents shall deviate from 13 to 16 avoirdupois ounces (see 3.3.1.1)
e. Whether the contractor is responsible for inspection (see 4.1)
f. Whether the contractor should use his own laboratory or other suitable facility (see 4.1)
g. Special inspection conditions (see 4.3)
h. The lot size (see 4.4.1)
i. Whether packing shall be Level A or commercial (see 5.2.1 and 5.2.2)
j. Additional marking (see 5.3.1)

6.3 Toxicological formulations. Material safety data sheet requirements are applicable and should be specified in the contract as required by DoD FAR 23.303.

6.4 Material Safety Data Sheets (MSDS). Contracting officers will identify those activities requiring copies of MSCSs. Additional Government mailing information is contained in Appendix B of FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.5 Subject term (key word) listing.

Aerosol
Alkyd
Properties
Testing
White

6.6 Net contents. The unit can will be known as a 1-pint can. Unless otherwise specified, it shall be understood that the weight of the contents may be from 13 to 15 avoirdupois ounces, at the discretion of the supplier, provided the total solids of the contents supplied is in accordance with 3.3.1.1. The purpose of this is to permit wide latitude in formulation while assuring uniform value of each unit based on paint solids. The net weight therefore need not be specified.

Preparing activity:
Navy — OS
(Project number 8010-N316)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-82747(OS)	2. DOCUMENT DATE (YYMMDD) 1988 JANUARY 27

3. DOCUMENT TITLE
COATING, POLYURETHANE CLEAR, SAFFLOWER OIL MODIFIED ALKYD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER	b. TELEPHONE (Include Area Code) (1) Commercial: 301-743-4358/4510 (2) AUTOVON: 364-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE. INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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